

CSCI 5423

Biologically-inspired Multi-Agent Systems

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (86.2 / 100)	62.6 / 100

63

CURRENT AVG SCORE
across 6 job roles

89

POTENTIAL SCORE
with recommended additions

+38

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Explores the principles and emergent properties of collective dynamics through computational modeling and theory. Focuses on multi-agent systems using insights from biology, like the self-assemblage of cells and insect colony behavior. Topics include designing swarm intelligence, networked agents, cellular computing and self-assembly, optimization, synchronization, and evolutionary computation. Uses cross-discipline research developments to practice applied techniques. Biology background is not required.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	86.2	100.0	+13.8	LOW
Machine Learning Engineer	68.9	97.5	+28.6	MEDIUM
AI Engineer	68.0	93.8	+25.8	MEDIUM
ML Engineer	56.8	82.1	+25.3	MEDIUM
Data Scientist	53.6	91.5	+37.8	HIGH
MLOps Engineer	42.0	70.4	+28.4	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Applied Scientist

86.2 → 100.0 (+13.8 pts)

discovery scientific interdisciplinary problems models technical drug data

Machine Learning Engineer

68.9 → 97.5 (+28.6 pts)

data sets learning analytics tools business mobile design

AI Engineer

68.0 → 93.8 (+25.8 pts)

data robotics models learning build software technical model

ML Engineer

56.8 → 82.1 (+25.3 pts)

media discovery data learning machine models model world

Data Scientist

53.6 → 91.5 (+37.8 pts)

data sets analytics social build people problems approaches

MLOps Engineer

42.0 → 70.4 (+28.4 pts)

data model learning machine media training tools models

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.